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 Studierichting: Informatica
 Oefeningenreeks 5L nr. 1.2

$$f(x) = \frac{1}{3}x^3$$

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$$S = 2\pi \int_0^1 \left| \frac{1}{3}x^3 \right| \sqrt{1 + \left(\left(\frac{1}{3}x^3 \right)' \right)^2} dx$$

$$\begin{aligned} & \int \left| \frac{1}{3}x^3 \right| \sqrt{1 + \left(\left(\frac{1}{3}x^3 \right)' \right)^2} dx \\ &= \frac{1}{3} \int |x^3| \sqrt{1 + x^4} dx \end{aligned}$$

Substitutie: $t = 1 + x^4 \Rightarrow dt = 4x^3 dx$

$$= \frac{1}{12} \int \sqrt{t} dt$$

$$= \frac{2t^{\frac{3}{2}}}{36} + c = \frac{(1 + x^4)^{\frac{3}{2}}}{18} + c$$

$$\Rightarrow S = 2\pi \left[\frac{(1 + x^4)^{\frac{3}{2}}}{18} \right]_0^1 = 2\pi \left(\frac{2^{\frac{3}{2}}}{18} - \frac{1}{18} \right) = \frac{2\pi\sqrt{2} - \pi}{9}$$

References